DATA WAREHOUSING AND MINING LABORATORY

Experiment No. 6: Generate Frequent Patterns Using FP-Growth Algorithm and Compare with Apriori

Aim: To generate frequent patterns using the FP-Growth algorithm in WEKA and compare its performance and output with the Apriori algorithm.

Objectives: After successful completion of this experiment, students will be able to:

1. Understand the concept of frequent pattern mining.
2. Understand the FP-Growth algorithm.
3. Understand the limitations of the Apriori algorithm.
4. Generate frequent patterns using FP-Growth in WEKA.
5. Compare FP-Growth and Apriori algorithms.
6. Interpret generated association rules.
7. Understand algorithm selection in data mining applications.

Importance of this Experiment:

Multiple algorithms can be used to discover frequent patterns and association rules from transactional datasets. Although Apriori and FP-Growth solve the same problem, they use different approaches. Understanding the strengths and limitations of each algorithm helps data analysts choose the most suitable technique for a given dataset. This experiment introduces students to algorithm comparison and performance evaluation, which are important skills in practical data mining projects.

Expected Outcome: After completing this experiment, students will be able to:

1. Apply FP-Growth using WEKA.
2. Generate frequent patterns from transactional datasets.
3. Compare FP-Growth and Apriori outputs.
4. Analyze support, confidence, and lift values.
5. Select appropriate algorithms for frequent pattern mining.

Facilities Required:

Hardware Requirements:

- Personal Computer / Laptop
- Intel i3 Processor or higher
- Minimum 4 GB RAM
- Internet Connection (Optional)

Software Requirements:

- Windows 10/11
- WEKA Tool
- Market Basket Dataset (market_basket_large.arff)

Prerequisite Knowledge: Students should be familiar with:

- Experiment No. 1: Installation and Exploration of WEKA Tool and Available Datasets
- Experiment No. 2: Create and Analyze ARFF Files Using WEKA
- Experiment No. 3: Data Cleaning and Missing Value Treatment
- Experiment No. 4: Data Transformation and Normalization
- Experiment No. 5: Generate Association Rules Using Apriori Algorithm
- Frequent Itemsets
- Support and Confidence

Theory:

1. Frequent Pattern Mining

Frequent pattern mining is the process of discovering recurring relationships among data items. For example, Milk, Bread appearing together in many transactions. Such recurring patterns are called frequent patterns. Frequent patterns form the foundation of:

- Recommendation Systems
- Market Basket Analysis
- Customer Behavior Analysis
- Retail Analytics

2. Review of Apriori Algorithm

In Experiment 5, students used the Apriori algorithm to generate association rules.

Apriori:

- Generates candidate itemsets.
- Performs multiple database scans.
- Generates frequent patterns using support and confidence thresholds.

3. Limitations of Apriori

Although Apriori is simple and easy to understand, it has several limitations.

1. Candidate Generation
    - Apriori repeatedly generates candidate itemsets.
2. Multiple Database Scans
    - The dataset must be scanned multiple times.
3. Increased Processing Time
    - Large datasets may require significant processing effort.

4. FP-Growth Algorithm

FP-Growth stands for: Frequent Pattern Growth. It is an efficient frequent pattern mining algorithm that avoids candidate generation. Instead of generating large numbers of candidate itemsets, FP-Growth uses a compressed data structure known as an FP-Tree. Advantages include:

- Faster execution
- Reduced database scans
- Better scalability
- Efficient mining of large datasets

5. FP-Tree Concept

FP-Growth stores transactions in a compact structure called an FP-Tree (Frequent Pattern Tree). An FP-Tree:

- Compresses transaction information.
- Preserves item associations.
- Eliminates repeated candidate generation.

This makes FP-Growth more efficient than Apriori when processing large datasets.

6. Apriori vs FP-Growth

Feature	Apriori	FP-Growth
Candidate Generation	Required	Not Required
Database Scans	Multiple	Fewer
Data Structure	Frequent Itemsets	FP-Tree
Efficiency	Moderate	Higher
Large Dataset Handling	Less Efficient	More Efficient
Ease of Understanding	Easier	Moderate

Why Experiment 6 Exists: Although Apriori and FP-Growth solve the same frequent pattern mining problem, FP-Growth is generally more efficient because it avoids repeated candidate generation. This experiment introduces students to the concept of: Algorithm Comparison and Algorithm Selection which are important skills in practical data mining projects.

Industry Insight: Modern retail analytics systems, e-commerce platforms, customer recommendation engines, and online shopping applications frequently use FP-Growth because it can process large transactional datasets more efficiently than Apriori.

Mini Real-World Example:

Online Shopping Platform

Transactions:

1. Mobile
2. Earphones
3. Power Bank

FP-Growth can identify patterns such as: Mobile → Earphones which can be used for product recommendations.

Food Recommendation System

Child	Preferred Foods
C1	Milk, Banana
C2	Milk, Apple
C3	Milk, Banana
C4	Banana, Apple

Possible frequent pattern: Milk + Banana

Such patterns can support personalized food recommendation systems.

Dataset Used in This Experiment:

Filename: market_basket_large.arff

Dataset Contents:

```
@relation marketbasket
@attribute Milk {t,f}
@attribute Bread {t,f}
@attribute Eggs {t,f}
@attribute Butter {t,f}
@attribute Tea {t,f}
@attribute Jam {t,f}
@attribute Cheese {t,f}
@data
t,t,f,f,t,f,f
t,f,t,f,f,f,f
f,t,f,t,t,f,f
t,t,t,f,f,f,f
t,t,f,t,f,f,f
f,t,f,f,t,t,f
t,f,f,f,t,f,t
f,t,f,f,f,t,f
t,t,f,f,f,f,t
f,f,t,t,t,f,f
t,t,t,f,t,f,f
t,t,f,t,f,t,f
f,t,f,t,t,f,t
t,f,t,f,f,t,f
t,t,f,f,t,f,t
```

Dataset Information:

- Relation Name: marketbasket
- Number of Attributes: 7

- Number of Transactions: 15
- Attribute Type: Nominal

Results Obtained During Experiment:

Apriori Output Summary:

Rules Generated: 20

Example Rule: Eggs=f ==> Bread=t

Confidence: 90%

Support: 9 Transactions

FP-Growth Output Summary:

Rules Generated: 27

Example Rule: Tea=f ==> Cheese=f

Confidence: 86%

Support: 6 Transactions

Experimental Comparison of Apriori and FP-Growth:

Parameter	Apriori	FP-Growth
Dataset Used	market_basket_large.arff	market_basket_large.arff
Transactions	15	15
Attributes	7	7
Rules Generated	20	27
Mining Strategy	Candidate Generation	FP-Tree
Example Rule	Eggs=f ⇒ Bread=t	Tea=f ⇒ Cheese=f
Confidence	90%	86%
Support (Example Rule)	9 Transactions	6 Transactions
Typical Usage	Small to Medium Datasets	Medium to Large Datasets

Understanding Generated Patterns:

Apriori Rule:

Eggs=f ==> Bread=t

Interpretation: Transactions not containing Eggs frequently contain Bread.

Confidence: 90%

Meaning: 90% of transactions that do not contain Eggs contain Bread.

FP-Growth Rule:

Tea=f ==> Cheese=f

Interpretation: Transactions not containing Tea frequently do not contain Cheese.

Confidence: 86%

Meaning: 86% of transactions that do not contain Tea also do not contain Cheese.

Procedure:

Activity 1: Load Dataset

Steps:

1. Open WEKA
2. Select Explorer
3. Open the Preprocess tab
4. Click Open File
5. Select market_basket_large.arff
6. Click Open

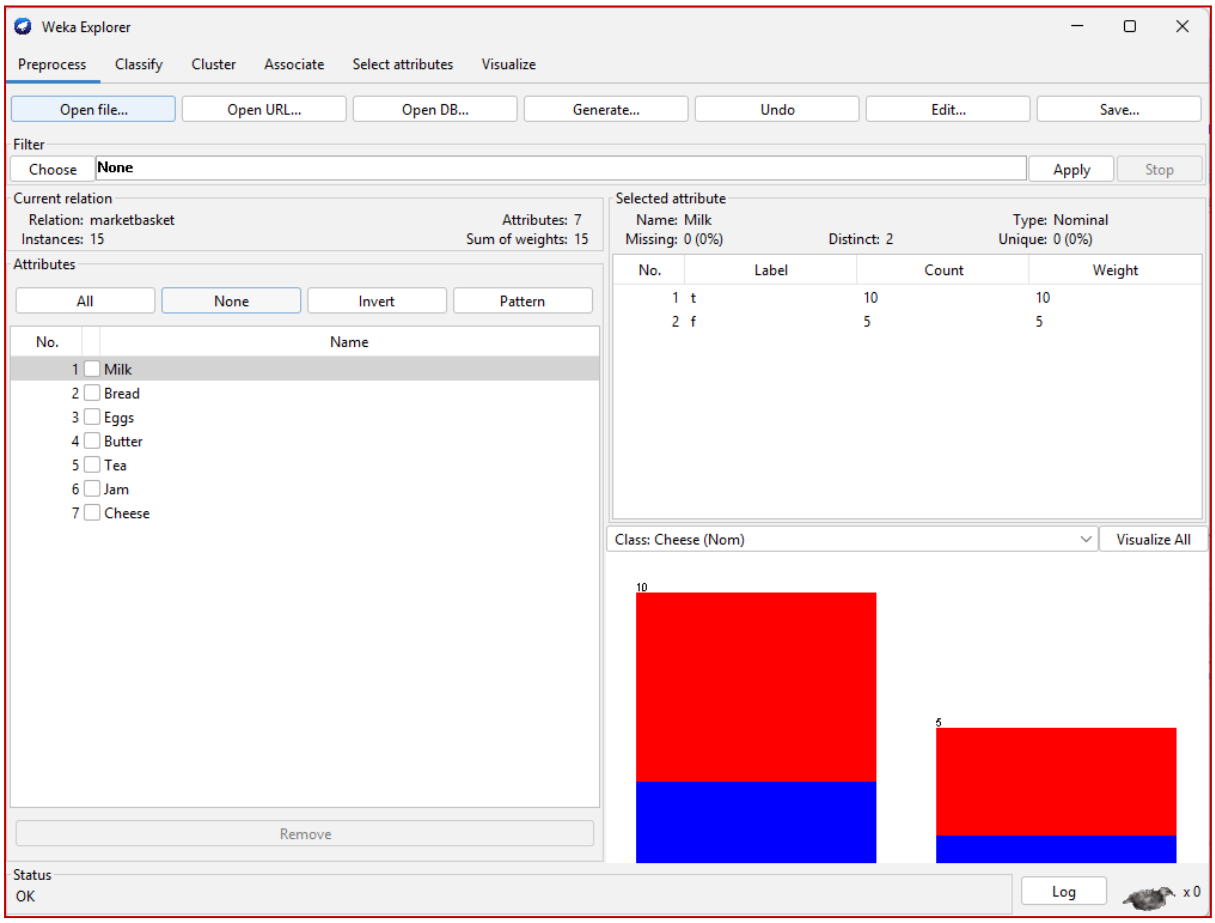


Figure 6.1: Market Basket Dataset Loaded in WEKA

Activity 2: Open Associate Tab

Step: Select: Associate tab.

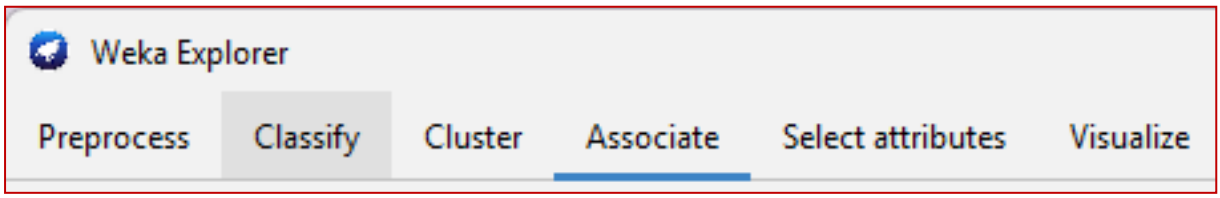


Figure 6.2: WEKA Associate Tab

Activity 3: Select FP-Growth Algorithm

Steps:

1. Click: Choose
2. Select: FP-Growth
3. Set:
   i. numRulesToFind = 20
   ii. minMetric = 0.7
   iii. positiveIndex = 2
4. Click OK.

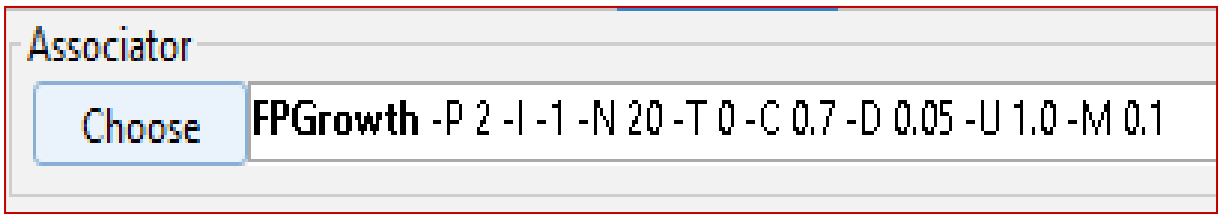


Figure 6.3: Selecting FP-Growth Algorithm

Activity 4: Generate Frequent Patterns

Steps:

1. Click: Start
2. Observe:
   - Frequent Patterns
   - Support Values
   - Confidence Values
   - Association Rules

Associator output

```
=== Run information ===

Scheme:       weka.associations.FPGrowth -P 2 -I -1 -N 20 -T 0 -C 0.7 -D 0.05 -U 1.0 -M 0.1
Relation:     marketbasket
Instances:    15
Attributes:   7
              Milk
              Bread
              Eggs
              Butter
              Tea
              Jam
              Cheese
=== Associator model (full training set) ===

FPGrowth found 27 rules (displaying top 20)

 1. [Butter=f, Milk=f]: 2 ==> [Cheese=f]: 2   <conf:(1)> lift:(1.36) lev:(0.04) conv:(0.53)
 2. [Tea=f, Bread=f]: 2 ==> [Cheese=f]: 2   <conf:(1)> lift:(1.36) lev:(0.04) conv:(0.53)
 3. [Butter=f, Milk=f]: 2 ==> [Eggs=f]: 2   <conf:(1)> lift:(1.5) lev:(0.04) conv:(0.67)
 4. [Tea=f, Bread=f]: 2 ==> [Butter=f]: 2   <conf:(1)> lift:(1.5) lev:(0.04) conv:(0.67)
 5. [Butter=f, Milk=f]: 2 ==> [Cheese=f, Eggs=f]: 2   <conf:(1)> lift:(2.5) lev:(0.08) conv:(1.2)
 6. [Cheese=f, Butter=f, Milk=f]: 2 ==> [Eggs=f]: 2   <conf:(1)> lift:(1.5) lev:(0.04) conv:(0.67)
 7. [Eggs=f, Butter=f, Milk=f]: 2 ==> [Cheese=f]: 2   <conf:(1)> lift:(1.36) lev:(0.04) conv:(0.53)
 8. [Cheese=f, Butter=f, Bread=f]: 2 ==> [Tea=f]: 2   <conf:(1)> lift:(2.14) lev:(0.07) conv:(1.07)
 9. [Tea=f, Bread=f]: 2 ==> [Cheese=f, Butter=f]: 2   <conf:(1)> lift:(2.14) lev:(0.07) conv:(1.07)
10. [Cheese=f, Tea=f, Bread=f]: 2 ==> [Butter=f]: 2   <conf:(1)> lift:(1.5) lev:(0.04) conv:(0.67)
11. [Butter=f, Tea=f, Bread=f]: 2 ==> [Cheese=f]: 2   <conf:(1)> lift:(1.36) lev:(0.04) conv:(0.53)
12. [Tea=f]: 7 ==> [Cheese=f]: 6   <conf:(0.86)> lift:(1.17) lev:(0.06) conv:(0.93)
13. [Milk=f]: 5 ==> [Cheese=f]: 4   <conf:(0.8)> lift:(1.09) lev:(0.02) conv:(0.67)
14. [Milk=f]: 5 ==> [Eggs=f]: 4   <conf:(0.8)> lift:(1.2) lev:(0.04) conv:(0.83)
15. [Butter=f, Tea=f]: 5 ==> [Cheese=f]: 4   <conf:(0.8)> lift:(1.09) lev:(0.02) conv:(0.67)
16. [Bread=f]: 4 ==> [Jam=f]: 3   <conf:(0.75)> lift:(1.02) lev:(0) conv:(0.53)
17. [Bread=f]: 4 ==> [Cheese=f]: 3   <conf:(0.75)> lift:(1.02) lev:(0) conv:(0.53)
18. [Bread=f]: 4 ==> [Butter=f]: 3   <conf:(0.75)> lift:(1.13) lev:(0.02) conv:(0.67)
19. [Jam=f, Tea=f]: 4 ==> [Cheese=f]: 3   <conf:(0.75)> lift:(1.02) lev:(0) conv:(0.53)
20. [Jam=f, Tea=f]: 4 ==> [Butter=f]: 3   <conf:(0.75)> lift:(1.13) lev:(0.02) conv:(0.67)
```

Figure 6.4: Generated Frequent Patterns Using FP-Growth

Activity 5: Compare with Apriori

Compare:

- Number of Rules
- Confidence Values
- Support Values
- Ease of Interpretation
- Mining Strategy

Record the observations.

Common Mistakes While Applying FP-Growth

1. Comparing algorithm outputs using different datasets.
2. Ignoring support and considering only confidence.
3. Assuming more rules always indicate better performance.
4. Confusing FP-Growth with Apriori.
5. Ignoring the practical usefulness of generated rules.

Common Discussion

1. Why was FP-Growth developed?
2. What are the limitations of Apriori?
3. What is an FP-Tree?
4. Why FP-Growth is considered efficient?
5. Which algorithm would be suitable for very large datasets?
6. Can two algorithms generate different rules from the same dataset?
7. What factors should be considered while evaluating association rules?

Student Activity:

Apply: Apriori and FP-Growth on the same dataset and compare:

- Number of Rules Generated
- Support Values
- Confidence Values
- Ease of Interpretation
- Strongest Rule Generated

Modify: Minimum Confidence and Number of Rules and observe:

- Number of generated rules
- Changes in support
- Changes in confidence
- Variation in discovered patterns

Observation Table:

PARAMETER	APRIORI	FP-GROWTH
Dataset Name		
Dataset File Name		
Number of Attributes		
Number of Transactions		
Rules Generated		
Strongest Rule		
Highest Confidence		
Mining Strategy		
Observation		

Result: Frequent patterns were successfully generated using the FP-Growth algorithm in WEKA. The results obtained using FP-Growth were compared with those generated by the Apriori algorithm. The experiment demonstrated the differences in mining strategies, generated patterns, and suitability of both algorithms for frequent pattern mining tasks.

Viva Questions:

1. What is FP-Growth?
2. Expand FP-Growth.
3. What is an FP-Tree?
4. Why was FP-Growth developed?
5. What are the limitations of Apriori?
6. Compare Apriori and FP-Growth.
7. What is frequent pattern mining?
8. What is candidate generation?
9. Why does FP-Growth avoid candidate generation?
10. Which algorithm is suitable for large datasets?
11. What is support?
12. What is confidence?

13. What is lift?
14. Why is algorithm comparison important?
15. Give two applications of FP-Growth.

Assignment Questions:

1. Explain the FP-Growth algorithm with a suitable example.
2. Compare Apriori and FP-Growth.
3. What is an FP-Tree and why is it used?
4. Explain the advantages of FP-Growth over Apriori.
5. Apply FP-Growth on a transactional dataset and analyze the generated patterns.
6. Explain the role of support and confidence in frequent pattern mining.
7. Discuss applications of FP-Growth in recommendation systems.

References:

1. Ian H. Witten, Eibe Frank and Mark A. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann.
2. Jiawei Han, Micheline Kamber and Jian Pei, *Data Mining: Concepts and Techniques*, Morgan Kaufmann.
3. WEKA Official Documentation & User Manual.

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